
EDUCATION

University of Oxford

- PhD in Statistical Machine Learning Oct 2024 – Apr 2028 (expected)
- Research Area: Diffusion Models, Markov Chain Monte Carlo, Gradient Estimation, Optimisation
- Supervisors: Yee Whye Teh & George Deligiannidis
- Scholarships: Google DeepMind PhD Scholarship

University of New South Wales

- Master of Mathematics Jun 2020 – Dec 2023
 - Master of Information Technology Jul 2017 – Jan 2020
 - Bachelor of Commerce/Science Feb 2011 – Dec 2015
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PUBLICATIONS & PREPRINTS

First or shared-first author:

- *Metropolis-Adjusted Diffusion Models*, **K. H. Lam***, T. Farghly*, C. Williams*, J. Yang, Y. W. Teh, A. Doucet, arXiv
- *Rao-Blackwellised Reparameterisation Gradients*, **K. H. Lam**, T. D. Bui, G. Deligiannidis, Y. W. Teh, Conference on Neural Information Processing Systems 2025, NeurIPS 2025
- *LegendreTron: Uprising Proper Multiclass Loss Learning*, **K. H. Lam**, C. Walder, S. Penev, R. Nock, International Conference on Machine Learning 2023, ICML 2023

Co-author:

- *Permutation-Invariant Spectral Learning via Dyson Diffusion*, T. Schwarz, C. Dieball, C. Kogler, **K. Lam**, R. Lambiotte, A. Doucet, A. Godec, G. Deligiannidis, arXiv
 - *Regularized Black-box Optimization Algorithms for Least-squares Problems*, Y. Liu, **K. H. Lam**, L. Roberts, IMA Journal of Numerical Analysis, IMAJNA 2025
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RESEARCH EXPERIENCE

Australian National University (ANU) – Research Intern/Assistant Nov 2022 – Oct 2024

- Research with Thang Bui on Variational Inference for Discrete Neural Networks

University of Technology Sydney (UTS) – Research Assistant Nov 2023 – Apr 2024

- Research with Roman Marchant and Hadi Afshar on Structure Learning for Bayesian Networks and LLMs

Australian National University (ANU) – Research Intern Nov 2023 – Jan 2024

- Research with Peter Vassiliou on Control Theory and Differential Geometry

Australian National University (ANU) – Research Intern Nov 2021 – Jan 2022

- Research with Lindon Roberts on Regularised Black-box Optimisation algorithms

Department of Defence (DSTG) – Research Cadet May 2021 – Nov 2021

- Research on Adversarial Machine Learning

University of Sydney (USYD) – Research Intern/Engineer Dec 2020 – May 2021

- Research on Manifold MCMC algorithms
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TEACHING

SC5 Advanced Simulation Methods – Teaching Assistant Jan 2026 – Mar 2026**SB2.2 Statistical Machine Learning** – Teaching Assistant Jan 2025 – Mar 2025**COMP9444 Neural Nets & Deep Learning** – Consultation Tutor Sep 2020 – Dec 2020

SKILLS

- Programming – Python, SQL, R, Matlab, C, C++
- Tooling – Git, Kubernetes, Docker, Docker Compose
- Python Libraries – PyTorch, JAX, Tensorflow, XGBoost, LightGBM, scikit-learn, pandas, NumPy, SciPy

INDUSTRY EXPERIENCE

QBE – Machine Learning Engineer	Jan 2024 – Sep 2024
- Automated Docker image deployments to JBFrog Artifactory with Azure Pipelines - Deployed an automated database backup service with Docker Compose and cron jobs	
Hivery – Machine Learning Engineer	Aug 2022 – Jan 2024
- Led MLOps adoption using Azure Databricks and MLflow for model tracking and versioning - Automated Docker image builds and registry pushes for ML models using GitLab CI/CD - Designed and deployed demand transfer models to optimise retail assortment	
Woolworths (WooliesX) – Senior Data Scientist	Jul 2018 – Oct 2019
- Implemented ML models for loyalty programs including Personalisation and Customer Lifetime Value - Illustrated effectiveness of ML-driven marketing through a randomised controlled trial and A/B testing	
Quantium – Data Analyst	Nov 2017 – May 2018
Identified home delivery expansion opportunities for a national retailer	
Commonwealth Bank – Quantitative Analyst	Jan 2017 – Oct 2017
Developed IRB models for regulatory capital by modelling probabilities of default	
AMP – Financial Risk Analyst	Jan 2016 – Dec 2016

PROFESSIONAL ACCREDITATIONS

Global Association of Risk Professionals – Financial Risk Manager	Sep 2021 – Present
Statistical Society of Australia – Graduate Statistician	Nov 2020 – Dec 2023
CFA Institute – Chartered Financial Analyst	Sep 2020 – June 2021